

PROBABILITY: SAMPLE SPACE & TREE DIAGRAMS

Fun with Chances and Choices!

Grade 6 Mathematics

What is Probability?

Probability tells us **how likely something is to happen!**

Everyday Examples:

- Tossing a coin — Will it be heads or tails?
- Rolling dice — Which number will come up?
- Picking a card — What colour will you get?

Probability helps us **guess what might happen next!**

Understanding Sample Space

Sample space is the list of **ALL possible outcomes**.

Example: Tossing a coin $\rightarrow$ { Heads, Tails }

Sample Space: Dice Roll

When you roll a standard 6-sided dice, the sample space is:

Dice Face	Dice Face	Dice Face	Dice Face	Dice Face	Dice Face
1	2	3	4	5	6

Outcomes: { 1, 2, 3, 4, 5, 6 } → 6 possible outcomes

Sample Space: Tossing Two Coins

Tossing **two coins** gives us:

Coin 1	Coin 2	Outcome
Heads	Heads	HH
Heads	Tails	HT
Tails	Heads	TH
Tails	Tails	TT

Total: 4 possible outcomes ($2 \times 2 = 4$)

Try It Yourself! — What's the Sample Space?

You pick a card from a set of 4 cards: ♥ Hearts, ♠ Spades, ♣ Clubs, ♦ Diamonds

What is the Sample Space?

Write down all possible outcomes in { } notation.

Answer: { ♥ Hearts, ♠ Spades, ♣ Clubs, ♦ Diamonds }

What is a Tree Diagram?

A **tree diagram** helps us list all possible outcomes step by step!

Why use it? To organise outcomes clearly and see all choices at once.

Think of it like **branches on a tree** — each branch shows a different choice or result!

■ Start → Branch 1 → Branch 2 → Final Outcome

Building a Tree Diagram: Two Coin Tosses

Step 1: First coin toss → Heads **or** Tails

Step 2: For each result, toss the second coin → Heads **or** Tails

Toss 1	Toss 2	Outcome
Heads (H)	Heads (H)	HH
	Tails (T)	HT
Tails (T)	Heads (H)	TH
	Tails (T)	TT

Total outcomes: 4 (2 choices × 2 choices)

Tree Diagram: Coin & Dice

Toss a **coin** (H or T) and roll a **dice** (1–6).

Total outcomes = 2 (coin) × 6 (dice) = 12 outcomes

Examples from the tree: H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6

Key Rule: Multiply the number of choices at each branch to get the total.

Using Tree Diagrams to Find Probability

Once we have the tree diagram, we can calculate probability:

$$P(\text{Event}) = \text{Favourable Outcomes} \div \text{Total Outcomes}$$

The answer is always between **0** (impossible) and **1** (certain).

It can also be written as a **fraction**, **decimal**, or **percentage**.

Practice Problem 1

You toss a coin and roll a dice.

What is the probability of getting Tails and a 4?

Step 1 — Total outcomes: $2 \times 6 = 12$

Step 2 — Favourable outcomes (T, 4): **1** (only one branch)

$$P(\text{Tails and 4}) = 1 \div 12 = 1/12$$

Practice Problem 2

You toss **two coins**.

$P(\text{exactly 1 Head}) = ?$

Step 1 — Total outcomes: $\{ HH, HT, TH, TT \} = 4$

Step 2 — Favourable outcomes (exactly 1 H): $HT, TH = 2$

$$P(\text{exactly 1 Head}) = 2 \div 4 = 1/2 = 50\%$$

Summary: What We Learned Today

■	Probability means chance — how likely an event is to happen.
■	Sample space is the complete list of all possible outcomes.
■	Tree diagrams help list outcomes step by step, branch by branch.
■	Probability formula: $\text{Favourable Outcomes} \div \text{Total Outcomes}$

Keep Exploring Probability!

Probability is all around us! Keep asking **"What can happen?"**

Try fun experiments at home:

- Coin toss games — record results over many tosses
- Dice rolls — how often do you roll a 6?
- Picking cards — predict and check!

The more you explore, the better you'll understand chances and choices.

Math is everywhere!

Thank You for Learning with Us!

Keep practising — you're doing great! ■